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EMMD: Question Bank on Unit-2 — Full Worked Solutions**Q1. Discrete Distribution — Basic Measures****Solution.** The probabilities must sum to 1:

$$k + 2k + 3k + 2k + k = 1 \Rightarrow 9k = 1 \Rightarrow k = \frac{1}{9}$$

So the distribution is

X	0	1	2	3	4
$P(X = x)$	$1/9$	$2/9$	$3/9$	$2/9$	$1/9$

 $P(X < 3)$:

$$P(X < 3) = P(0) + P(1) + P(2) = \frac{1}{9} + \frac{2}{9} + \frac{3}{9} = \frac{6}{9} = \frac{2}{3}$$

 $P(1 \leq X \leq 4)$:

$$P(1 \leq X \leq 4) = P(1) + P(2) + P(3) + P(4) = \frac{2}{9} + \frac{3}{9} + \frac{2}{9} + \frac{1}{9} = \frac{8}{9}$$

 $E(X)$:

$$E(X) = \sum xP(x) = 0 \cdot \frac{1}{9} + 1 \cdot \frac{2}{9} + 2 \cdot \frac{3}{9} + 3 \cdot \frac{2}{9} + 4 \cdot \frac{1}{9} = \frac{0 + 2 + 6 + 6 + 4}{9} = \frac{18}{9} = 2$$

 $V(X)$: First find $E(X^2)$:

$$E(X^2) = \sum x^2P(x) = 0 \cdot \frac{1}{9} + 1 \cdot \frac{2}{9} + 4 \cdot \frac{3}{9} + 9 \cdot \frac{2}{9} + 16 \cdot \frac{1}{9} = \frac{0 + 2 + 12 + 18 + 16}{9} = \frac{48}{9} = \frac{16}{3}$$

$$V(X) = E(X^2) - [E(X)]^2 = \frac{16}{3} - 4 = \frac{16 - 12}{3} = \frac{4}{3}$$

Final Answer: $k = \frac{1}{9}$, $P(X < 3) = \frac{2}{3}$, $P(1 \leq X \leq 4) = \frac{8}{9}$, $E(X) = 2$, $V(X) = \frac{4}{3}$.

Q2. Discrete Distribution — Finding k and Probabilities**Solution.**

$$X : 1, 2, 3, 4, 5, 6, 7, \quad P(x) : k, 2k, 3k, k^2, k^2 + k, 2k^2, 4k^2$$

(i) Summing all probabilities to 1:

$$k + 2k + 3k + k^2 + (k^2 + k) + 2k^2 + 4k^2 = 1$$

Collecting the plain k terms: $k+2k+3k+k=7k$. Collecting the k^2 terms: $k^2+k^2+2k^2+4k^2=8k^2$. This gives

$$8k^2 + 7k - 1 = 0$$

Solving the quadratic: $k = \frac{-7 \pm \sqrt{49+32}}{16} = \frac{-7 \pm 9}{16}$, giving $k = \frac{1}{8}$ or $k = -1$. Since a probability cannot be negative, $k = \frac{1}{8}$.

With $k = \frac{1}{8}$: $k^2 = \frac{1}{64}$, so the table becomes

X	1	2	3	4	5	6	7
$P(X=x)$	$\frac{1}{8}$	$\frac{2}{8}$	$\frac{3}{8}$	$\frac{1}{64}$	$\frac{9}{64}$	$\frac{2}{64}$	$\frac{4}{64}$

(Here $P(5) = k^2 + k = \frac{1}{64} + \frac{8}{64} = \frac{9}{64}$.)

(ii) $P(X > 5)$:

$$P(X > 5) = P(6) + P(7) = \frac{2}{64} + \frac{4}{64} = \frac{6}{64} = \frac{3}{32}$$

(iii) $P(1 \leq X \leq 5)$:

$$P(1 \leq X \leq 5) = P(1) + P(2) + P(3) + P(4) + P(5) = \frac{8}{64} + \frac{16}{64} + \frac{24}{64} + \frac{1}{64} + \frac{9}{64} = \frac{58}{64} = \frac{29}{32}$$

Final Answer: $k = \frac{1}{8}$; $P(X > 5) = \frac{3}{32}$; $P(1 \leq X \leq 5) = \frac{29}{32}$.

Q3. Continuous Distribution — $f(x) = kx(2-x)$

Solution. (i) Total probability is 1:

$$\int_0^2 kx(2-x) dx = 1$$

$$k \int_0^2 (2x - x^2) dx = k \left[x^2 - \frac{x^3}{3} \right]_0^2 = k \left(4 - \frac{8}{3} \right) = k \cdot \frac{4}{3} = 1 \Rightarrow k = \frac{3}{4}$$

(ii) **Mean:**

$$\begin{aligned} E(X) &= \int_0^2 x \cdot \frac{3}{4}x(2-x) dx = \frac{3}{4} \int_0^2 (2x^2 - x^3) dx = \frac{3}{4} \left[\frac{2x^3}{3} - \frac{x^4}{4} \right]_0^2 \\ &= \frac{3}{4} \left(\frac{16}{3} - 4 \right) = \frac{3}{4} \cdot \frac{4}{3} = 1 \end{aligned}$$

(iii) **Standard deviation:** First compute $E(X^2)$:

$$\begin{aligned} E(X^2) &= \int_0^2 x^2 \cdot \frac{3}{4}x(2-x) dx = \frac{3}{4} \int_0^2 (2x^3 - x^4) dx = \frac{3}{4} \left[\frac{x^4}{2} - \frac{x^5}{5} \right]_0^2 \\ &= \frac{3}{4} \left(8 - \frac{32}{5} \right) = \frac{3}{4} \cdot \frac{8}{5} = \frac{6}{5} \end{aligned}$$

$$V(X) = E(X^2) - [E(X)]^2 = \frac{6}{5} - 1 = \frac{1}{5}$$

$$\sigma = \sqrt{\frac{1}{5}} = \frac{1}{\sqrt{5}} \approx 0.4472$$

Final Answer: $k = \frac{3}{4}$, $E(X) = 1$, $\sigma = \frac{1}{\sqrt{5}} \approx 0.4472$.

Q4. Continuous Distribution — $f(x) = \lambda x e^{-x}$

Solution. (i) Normalize the density using $\int_0^\infty x e^{-x} dx = 1$ (a standard Gamma integral, $\Gamma(2) = 1! = 1$):

$$\int_0^\infty \lambda x e^{-x} dx = \lambda \cdot 1 = 1 \Rightarrow \lambda = 1$$

So $f(x) = x e^{-x}$.

(ii) We need

$$P(2 < X < 5) = \int_2^5 x e^{-x} dx$$

Integrate by parts with $u = x$, $dv = e^{-x} dx \Rightarrow du = dx$, $v = -e^{-x}$:

$$\int x e^{-x} dx = -x e^{-x} + \int e^{-x} dx = -x e^{-x} - e^{-x} = -(x+1)e^{-x}$$

So

$$\int_2^5 x e^{-x} dx = \left[-(x+1)e^{-x} \right]_2^5 = -(6)e^{-5} - (-(3)e^{-2}) = 3e^{-2} - 6e^{-5}$$

Numerically: $3e^{-2} \approx 0.40601$, $6e^{-5} \approx 0.04043$, so

$$P(2 < X < 5) \approx 0.40601 - 0.04043 = 0.36558 \approx 0.3656$$

Final Answer: $\lambda = 1$; $P(2 < X < 5) = 3e^{-2} - 6e^{-5} \approx 0.3656$.

Q5. Binomial — Defective Bolts

Solution. Here $X \sim B(n, p)$ with $n = 8$, $p = 0.2$ (defective), $q = 0.8$. The pmf is $P(X = x) = \binom{8}{x} (0.2)^x (0.8)^{8-x}$.

(i) **Exactly 3 defective:**

$$P(X = 3) = \binom{8}{3} (0.2)^3 (0.8)^5 = 56 \times 0.008 \times 0.32768 = 0.14680064$$

(ii) **No bolt defective:**

$$P(X = 0) = (0.8)^8 = 0.16777216$$

(iii) **At most 2 defective:**

$$P(X \leq 2) = P(0) + P(1) + P(2)$$

$$P(1) = \binom{8}{1} (0.2)(0.8)^7 = 8(0.2)(0.2097152) = 0.33554432$$

$$P(2) = \binom{8}{2} (0.2)^2 (0.8)^6 = 28(0.04)(0.262144) = 0.29360128$$

$$P(X \leq 2) = 0.16777216 + 0.33554432 + 0.29360128 = 0.79691776$$

(iv) **At least 6 defective:**

$$P(X \geq 6) = P(6) + P(7) + P(8)$$

$$P(6) = \binom{8}{6} (0.2)^6 (0.8)^2 = 28(0.000064)(0.64) = 0.00114688$$

$$P(7) = \binom{8}{7} (0.2)^7 (0.8)^1 = 8(0.0000128)(0.8) = 0.00008192$$

$$P(8) = (0.2)^8 = 0.00000256$$

$$P(X \geq 6) = 0.00114688 + 0.00008192 + 0.00000256 = 0.00123136$$

Final Answer: (i) 0.14680064; (ii) 0.16777216; (iii) 0.79691776; (iv) 0.00123136.

Q6. Binomial — Doublets in Dice Throws

Solution. A doublet has probability $p = \frac{1}{6}$ per throw (i.e. both dice show the same face), so $q = \frac{5}{6}$. Here $X \sim B(20, \frac{1}{6})$.

(i) **Exactly 4 successes:**

$$P(X = 4) = \binom{20}{4} \left(\frac{1}{6}\right)^4 \left(\frac{5}{6}\right)^{16}$$

Since $\binom{20}{4} = 4845$, and evaluating numerically:

$$P(X = 4) \approx 4845 \times 0.00077160 \times 0.05407 \approx 0.202204$$

(ii) **At most 18 successes:** Use the complementary event, since $P(X \geq 19)$ has only two terms:

$$P(X \leq 18) = 1 - P(X = 19) - P(X = 20)$$

$$P(X = 19) = \binom{20}{19} \left(\frac{1}{6}\right)^{19} \left(\frac{5}{6}\right) = 20 \left(\frac{1}{6}\right)^{19} \left(\frac{5}{6}\right)$$

$$P(X = 20) = \left(\frac{1}{6}\right)^{20}$$

Both terms are astronomically small (of order 10^{-14}), so

$$P(X \leq 18) \approx 1 - 2.8 \times 10^{-14} \approx 0.999999999999972$$

(iii) **No success:**

$$P(X = 0) = \left(\frac{5}{6}\right)^{20} \approx 0.0260841$$

Final Answer: (i) ≈ 0.202204 ; (ii) $\approx 0.999999999999972$; (iii) ≈ 0.0260841 .

Q7. Binomial — Finding the Mode from Mean and Variance**Solution.** For $X \sim B(n, p)$: mean = np and variance = npq where $q = 1 - p$.Given $np = 18$ and $npq = 9$:

$$q = \frac{npq}{np} = \frac{9}{18} = \frac{1}{2} \Rightarrow p = 1 - q = \frac{1}{2}$$

Then

$$n = \frac{np}{p} = \frac{18}{1/2} = 36$$

Mode: For a binomial distribution the mode is the integer part of $(n + 1)p$ (or, when $(n + 1)p$ is itself an integer, both $(n + 1)p$ and $(n + 1)p - 1$ are modes).

$$(n + 1)p = 37 \times \frac{1}{2} = 18.5$$

Since 18.5 is not an integer, the mode is $\lfloor 18.5 \rfloor = 18$.
Final Answer: $n = 36$, $p = \frac{1}{2}$, and the mode is 18.
Q8. Poisson — Phone Calls**Solution.** Calls arrive at rate $\lambda = 5$ per hour, so $X \sim \text{Poisson}(5)$ with $P(X = x) = \frac{e^{-\lambda} \lambda^x}{x!}$.

$$P(X \leq 2) = P(0) + P(1) + P(2) = e^{-5} \left(1 + 5 + \frac{5^2}{2!} \right) = e^{-5} (1 + 5 + 12.5) = 18.5 e^{-5}$$

Since $e^{-5} \approx 0.0067379$:

$$P(X \leq 2) \approx 18.5 \times 0.0067379 \approx 0.124652$$

Final Answer: $P(X \leq 2) = e^{-5} \left(1 + 5 + \frac{25}{2} \right) \approx 0.124652$.
Q9. Poisson Approximation — Bad Reactions**Solution.** Here $n = 2000$ is large and $p = 0.001$ is small, so we approximate the binomial by a Poisson distribution with

$$\lambda = np = 2000 \times 0.001 = 2$$

(i) Exactly 3 suffer a reaction:

$$P(X = 3) = \frac{e^{-2} 2^3}{3!} = \frac{8e^{-2}}{6} = \frac{4}{3} e^{-2} \approx \frac{4}{3} (0.135335) \approx 0.180447$$

(ii) More than 2 suffer a reaction: Use the complement,

$$P(X > 2) = 1 - P(X \leq 2) = 1 - e^{-2} \left(1 + 2 + \frac{2^2}{2!} \right) = 1 - e^{-2} (1 + 2 + 2) = 1 - 5e^{-2}$$

$$P(X > 2) \approx 1 - 5(0.135335) \approx 1 - 0.676676 \approx 0.323324$$

Final Answer: With $\lambda = np = 2$: (i) $P(X = 3) = \frac{4}{3} e^{-2} \approx 0.180447$; (ii) $P(X > 2) = 1 - 5e^{-2} \approx 0.323324$.

Q10. Fitting a Poisson Distribution (Data Set 1)**Solution.**

$$N = \sum f = 122 + 60 + 15 + 2 + 1 = 200$$

$$\sum fx = 0(122) + 1(60) + 2(15) + 3(2) + 4(1) = 0 + 60 + 30 + 6 + 4 = 100$$

$$\lambda = \bar{x} = \frac{\sum fx}{N} = \frac{100}{200} = 0.5$$

Theoretical (fitted) frequency for each x is $N \cdot P(X = x) = N \cdot \frac{e^{-\lambda} \lambda^x}{x!}$, with $e^{-0.5} \approx 0.606531$.

x	$P(X = x) = \frac{e^{-0.5}(0.5)^x}{x!}$	Theoretical freq. = $200P(x)$	Rounded
0	0.606531	121.306	121
1	0.303265	60.653	61
2	0.075816	15.163	15
3	0.012636	2.527	3
4	0.001580	0.316	0

Final Answer: $\lambda = 0.5$. Theoretical frequencies (rounded) for $x = 0, 1, 2, 3, 4$: 121, 61, 15, 3, 0.

Q11. Fitting a Poisson Distribution (Data Set 2)**Solution.**

$$N = \sum f = 109 + 65 + 22 + 3 + 1 = 200$$

$$\sum fx = 0(109) + 1(65) + 2(22) + 3(3) + 4(1) = 0 + 65 + 44 + 9 + 4 = 122$$

$$\lambda = \bar{x} = \frac{\sum fx}{N} = \frac{122}{200} = 0.61$$

With $e^{-0.61} \approx 0.543351$, the fitted (theoretical) frequencies = $200 \times \frac{e^{-0.61}(0.61)^x}{x!}$ are:

x	$P(X = x)$	Theoretical freq.	Rounded
0	0.543351	108.670	109
1	0.331444	66.289	66
2	0.101090	20.218	20
3	0.020555	4.111	4
4	0.003135	0.627	1

Final Answer: $\lambda = 0.61$. Theoretical frequencies (rounded) for $x = 0, 1, 2, 3, 4$: 109, 66, 20, 4, 1.

Q12. Poisson — Misprints per Page**Solution.** Here $\lambda = 1.2$, so $X \sim \text{Poisson}(1.2)$.**(i) No misprint:**

$$P(X = 0) = e^{-1.2} \approx 0.301194$$

(ii) Exactly 2 misprints:

$$P(X = 2) = \frac{e^{-1.2}(1.2)^2}{2!} = \frac{1.44}{2}e^{-1.2} = 0.72e^{-1.2} \approx 0.216860$$

(iii) At least one misprint: Use the complement of “no misprint”:

$$P(X \geq 1) = 1 - P(X = 0) = 1 - e^{-1.2} \approx 0.698806$$

Final Answer: (i) $e^{-1.2} \approx 0.301194$; (ii) $0.72e^{-1.2} \approx 0.216860$; (iii) $1 - e^{-1.2} \approx 0.698806$.
Q13. Geometric Distribution — Defective Bulbs

Solution. The probability a bulb is defective is $p = \frac{3}{60} = 0.05$, so $q = 0.95$. Let X be the trial number of the first defective bulb; then X follows a geometric distribution with pmf $P(X = x) = q^{x-1}p$.

Probability first defective is the 5th bulb:

$$P(X = 5) = q^4p = (0.95)^4(0.05)$$

Since $(0.95)^4 \approx 0.81450625$:

$$P(X = 5) \approx 0.81450625 \times 0.05 \approx 0.0407253$$

Mean, variance, and SD of a geometric distribution (number of trials to first success) are

$$E(X) = \frac{1}{p}, \quad V(X) = \frac{q}{p^2}$$

$$E(X) = \frac{1}{0.05} = 20$$

$$V(X) = \frac{0.95}{(0.05)^2} = \frac{0.95}{0.0025} = 380$$

$$\sigma = \sqrt{380} \approx 19.4936$$

Final Answer: $P(X = 5) = (0.95)^4(0.05) \approx 0.0407253$; mean = 20; variance = 380; SD = $\sqrt{380} \approx 19.4936$.

Q14. Geometric Distribution — Basketball Shots

Solution. Here $p = 0.7$ (success probability per shot), $q = 0.3$; X is the attempt number of the first successful basket, so $P(X = x) = q^{x-1}p$.

Probability the first basket is on the 4th attempt:

$$P(X = 4) = q^3p = (0.3)^3(0.7) = 0.027 \times 0.7 = 0.0189$$

Mean, variance, SD:

$$E(X) = \frac{1}{p} = \frac{1}{0.7} = \frac{10}{7} \approx 1.4286$$

$$V(X) = \frac{q}{p^2} = \frac{0.3}{0.49} = \frac{30}{49} \approx 0.6122$$

$$\sigma = \sqrt{\frac{30}{49}} = \frac{\sqrt{30}}{7} \approx 0.7825$$

Why the mode is always 1: For the geometric pmf $P(X = x) = q^{x-1}p$ with $0 < q < 1$,

$$\frac{P(X = x + 1)}{P(X = x)} = \frac{q^x p}{q^{x-1} p} = q < 1$$

so each successive probability is smaller than the one before it: $P(X = 1) > P(X = 2) > P(X = 3) > \dots$. The probability is therefore largest at $x = 1$, making the mode always equal to 1, regardless of the value of p .

Final Answer: $P(X = 4) = 0.0189$; mean $= \frac{10}{7} \approx 1.4286$; variance $= \frac{30}{49} \approx 0.6122$; SD $= \frac{\sqrt{30}}{7} \approx 0.7825$. The mode is always 1 because $P(X = x)$ is strictly decreasing in x .

Q15. Normal Distribution — Aptitude Test Scores

Solution. Let $X \sim N(\mu = 45, \sigma = 25)$, and use the standard score $z = \frac{x - \mu}{\sigma}$.

(i) Number scoring more than 60:

$$z = \frac{60 - 45}{25} = \frac{15}{25} = 0.6$$

The area between the mean and $z = 0.6$ is given as $A(0.6) = 0.2257$, so

$$P(X > 60) = 0.5 - A(0.6) = 0.5 - 0.2257 = 0.2743$$

Expected number of candidates:

$$1000 \times 0.2743 \approx 274$$

(ii) Number scoring between 30 and 60:

$$z_{30} = \frac{30 - 45}{25} = -0.6, \quad z_{60} = \frac{60 - 45}{25} = 0.6$$

Both bounds are equidistant from the mean, so

$$P(30 < X < 60) = A(0.6) + A(0.6) = 2(0.2257) = 0.4514$$

Expected number of candidates:

$$1000 \times 0.4514 \approx 451$$

Final Answer: (i) ≈ 274 candidates score above 60; (ii) ≈ 451 candidates score between 30 and 60.

Q16. Normal Distribution — Test Scores

Solution. Let $X \sim N(\mu = 14, \sigma = 2.5)$.

(i) **Number scoring between 12 and 15:**

$$z_{12} = \frac{12 - 14}{2.5} = -0.8, \quad z_{15} = \frac{15 - 14}{2.5} = 0.4$$

$$P(12 < X < 15) = A(0.8) + A(0.4) = 0.2881 + 0.1554 = 0.4435$$

Expected number of students:

$$1000 \times 0.4435 \approx 444$$

(ii) **Number scoring less than 8:**

$$z_8 = \frac{8 - 14}{2.5} = -2.4$$

$$P(X < 8) = 0.5 - A(2.4) = 0.5 - 0.4918 = 0.0082$$

Expected number of students:

$$1000 \times 0.0082 \approx 8$$

Final Answer: (i) ≈ 444 students score between 12 and 15; (ii) ≈ 8 students score less than 8.

Q17. Normal Distribution — Student Weights

Solution. Let $X \sim N(\mu = 50, \sigma = 5)$.

(i) **Number with weight less than 45 kg:**

$$z = \frac{45 - 50}{5} = -1$$

$$P(X < 45) = 0.5 - A(1) = 0.5 - 0.3413 = 0.1587$$

Expected number of students:

$$4000 \times 0.1587 \approx 635$$

(ii) **Number with weight between 45 kg and 60 kg:**

$$z_{45} = -1, \quad z_{60} = \frac{60 - 50}{5} = 2$$

$$P(45 < X < 60) = A(1) + A(2) = 0.3413 + 0.4772 = 0.8185$$

Expected number of students:

$$4000 \times 0.8185 \approx 3274$$

Final Answer: (i) ≈ 635 students weigh less than 45 kg; (ii) ≈ 3274 students weigh between 45 kg and 60 kg.

Q18. Normal Distribution — LED Bulb Lifetimes

Solution. Let $X \sim N(\mu = 1000, \sigma = 200)$ represent the bulb's lifetime in hours. We need the number of bulbs failing within 700 hours, i.e., $P(X < 700)$.

$$z = \frac{700 - 1000}{200} = \frac{-300}{200} = -1.5$$

$$P(X < 700) = 0.5 - A(1.5) = 0.5 - 0.4332 = 0.0668$$

Expected number of bulbs (out of 2000) failing within 700 hours:

$$2000 \times 0.0668 \approx 134$$

Final Answer: Approximately 134 LED bulbs are expected to fail within 700 hours.